Elementary Analysis

Section 31: Taylor's Theorem

Exercises 31.1–31.12, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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31 Taylor's theorem

A Taylor polynomial matches a function's derivatives at its center. Matching those derivatives does not by itself show that the infinite series represents the function. The missing step is to show that the remainder tends to zero. With Ross's convention the polynomial in R_n stops at degree $n-1$:

$$R_n(x) = f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!}(x-c)^k.$$

Taylor's theorem bounds this error using the nth derivative at a point between c and x. Later exercises use Newton's method, which replaces a curve by its tangent line and uses that line's zero as the next approximation.

31.1 Exercise 31.1: The cosine series and its error

What we must show. Find the Taylor series at 0 and prove that it equals $\cos x$ everywhere.

Compute the coefficients.

Successive derivatives of cosine are $\cos x, -\sin x, -\cos x, \sin x$, then repeat. At zero, the odd derivatives vanish and the $2k$th derivative is $(-1)^k$. Thus the Taylor series is

$$\sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots.$$

Control the error, rather than assuming the series is correct.

Every derivative of cosine has absolute value at most 1. Taylor's theorem therefore gives $|R_n(x)| \leq |x|^n / n!$ for each fixed real x. This bound tends to zero: if $a_n = |x|^n / n!$, then $a_{n+1}/a_n = |x| / (n+1) \leq 1/2$ eventually (and $x = 0$ is immediate). The tail is consequently bounded by a geometric sequence tending to zero.

Conclusion. The displayed series equals $\cos x$ for every $x \in \mathbb{R}$.

31.2 Exercise 31.2: The hyperbolic sine and cosine series

What we must show. Obtain series at zero for $\sinh x = (e^x - e^{-x})/2$ and $\cosh x = (e^x + e^{-x})/2$, valid on all of $\mathbb{R}$.

Add and subtract the exponential series.

For each real x, the exponential series is absolutely convergent: $e^x = \sum_{k\geq 0} x^k/k!$ and $e^{-x} = \sum_{k\geq 0}(-1)^k x^k/k!$. Subtracting cancels the even terms; adding cancels the odd terms. Hence

$$\sinh x = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}, \qquad \cosh x = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}.$$

Termwise addition here is legitimate because it is addition of convergent series at a fixed x.

Check that these really are Taylor coefficients.

Differentiation interchanges sinh and cosh. Their values at zero are 0 and 1, respectively. This gives exactly the displayed coefficients. Alternatively, on $[-M, M]$ all derivatives of either function are bounded by e^M, so Taylor's bound is $e^M |x|^n / n! \to 0$. For any given x, choose $M > |x|$; this proves validity at that x.

Conclusion. Both series represent their functions at every real number.

31.3 Exercise 31.3: Why the common derivative bound is unavailable

What we must show. Explain the theorem choice for $f(x) = \log(1 + x)$ in Example 2.

Inspect the hypothesis of Corollary 31.4.

That corollary requires a single constant bounding *all* derivatives on the interval in question. But

$$f^{(n)}(x) = (-1)^{n+1}(n-1)!(1+x)^{-n}, \qquad \left|f^{(n)}(0)\right| = (n-1)!.$$

Even at the center 0 these numbers have no common finite bound. No interval containing zero can satisfy the corollary's hypothesis.

Use the more precise remainder instead.

For evaluating at $x = 1$, Taylor's theorem gives a $y_n \in (0, 1)$ with

$$|R_n(1)| = \frac{1}{n(1+y_n)^n} \leq \frac{1}{n} \longrightarrow 0.$$

The factorial growth of the derivative is canceled by the $n!$ in the remainder. The corollary's stronger hypothesis would miss that cancellation.

Conclusion. Theorem 31.3 applies directly; the common-bound hypothesis of the corollary fails.

31.4 Exercise 31.4: Smooth cutoffs and a smooth step

What we must show. Construct the four requested infinitely differentiable functions.

Plan. Start from the smooth function in Example 3,

$$\phi(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \leq 0. \end{cases}$$

It is important that all its derivatives at zero vanish; this is what makes the joining point smooth, despite the piecewise formula.

(a) Move the start of the positive part to a.

Set $f_a(x) = \phi(x - a)$. A translation preserves smoothness. If $x \leq a$ its argument is nonpositive, so $f_a(x) = 0$; if $x > a$, its argument is positive and $f_a(x) > 0$.

(b) Reverse the direction and move the endpoint to b.

Set $g_b(x) = \phi(b - x)$. The affine change of variable preserves smoothness. This is zero for $x \geq b$ and strictly positive for $x < b$.

(c) Require both positivity conditions.

Set $h_{a,b}(x) = f_a(x)g_b(x)$. Products of smooth functions are smooth. Both factors are positive exactly when $a < x < b$. Outside this interval at least one factor is zero, so the product is zero.

(d) Turn the two cutoffs into a step.

Set

$$h^*_{a,b}(x) = \frac{f_a(x)}{f_a(x) + g_b(x)}.$$

The denominator never vanishes. Indeed, if $x \le a$ then $x < b$ and $g_b(x) > 0$; if $x \ge b$ then $x > a$ and $f_a(x) > 0$; between a and b both are positive. The permitted quotient rule for smooth functions therefore applies. For $x \le a$ the quotient is 0; for $x \ge b$ its numerator equals its denominator, so it is 1.

Conclusion. These four explicit formulas have all the required properties, including smoothness at a and b.

31.5 Exercise 31.5: A nonzero smooth function with zero Taylor series

What we must show. For $g(x) = e^{-1/x^2}$ when $x \neq 0$, $g(0) = 0$, find all derivatives at zero and compare g with its Taylor series.

(a) Recognize a smooth composition.

Using ϕ from Example 3, we have $g(x) = \phi(x^2)$ for every x. Since both ϕ and $x \mapsto x^2$ are smooth, so is g. Repeated chain and product rules show that each positive-order derivative of this composition is a finite sum of polynomial factors times $\phi^{(j)}(x^2)$, for integers $j \ge 1$. At zero every such term vanishes, because $\phi^{(j)}(0) = 0$. Also $g(0) = 0$. Thus $g^{(n)}(0) = 0$ for all $n = 0, 1, 2, \ldots$.

(b) Compare the function and the series point by point.

Every coefficient $g^{(n)}(0)/n!$ is zero, so the Taylor series is the identically zero series. It equals g at 0. For $x \neq 0$, $e^{-1/x^2} > 0$, so it cannot equal $g(x)$ there.

Conclusion. Smoothness alone does not imply agreement with a Taylor series, even arbitrarily close to its center.

31.6 Exercise 31.6: A telescoping proof of Taylor's theorem

What we must show. Verify the older proof using its auxiliary function F. Fix $x > 0$. As in the theorem's proof, define M by

$$f(x) = \sum_{k=0}^{n-1} \frac{x^k}{k!} f^{(k)}(0) + \frac{Mx^n}{n!}.$$

(a) Differentiate and cancel adjacent terms.

Write the auxiliary function as

$$F(t) = \sum_{k=0}^{n-1} \frac{(x-t)^k}{k!} f^{(k)}(t) + M\frac{(x-t)^n}{n!}, \qquad 0 \le t \le x.$$

For $k \geq 1$, the product rule gives

$$\frac{d}{dt}\left(\frac{(x-t)^k}{k!}f^{(k)}(t)\right) = -\frac{(x-t)^{k-1}}{(k-1)!}f^{(k)}(t) + \frac{(x-t)^k}{k!}f^{(k+1)}(t).$$

The negative term cancels the positive term from index $k-1$. Only the last positive term and the derivative of the M term remain:

$$F'(t) = \frac{(x-t)^{n-1}}{(n-1)!}(f^{(n)}(t) - M).$$

All needed derivatives exist on an open interval containing $[0, x]$; in particular F is continuous on the closed interval and differentiable in its interior. For $n = 1$ the formula follows directly from $F(t) = f(t) + M(x-t)$.

(b) Evaluate the endpoints.

The definition of M says $F(0) = f(x)$. At $t = x$ all positive powers of $x-t$ vanish, leaving $F(x) = f(x)$. Thus $F(0) = F(x)$.

(c) Apply Rolle's theorem.

There is $y \in (0, x)$ with $F'(y) = 0$. Since $x - y > 0$, the prefactor in the derivative formula is nonzero; hence $f^{(n)}(y) = M$. Substituting into the defining equation for M yields Taylor's remainder.

Conclusion. The auxiliary function proves the required mean-value expression for the remainder.

31.7 Exercise 31.7: Newton's method for the square root of two

What we must show. Recover the recursion from Exercise 9.5. For $f(x) = x^2 - 2$, we have $f'(x) = 2x$. Newton's update at a nonzero t_n is

$$t_{n+1} = t_n - \frac{t_n^2 - 2}{2t_n} = \frac{2t_n^2 - t_n^2 + 2}{2t_n} = \frac{t_n^2 + 2}{2t_n}.$$

With initial point $t_1 = 1$ this is exactly the earlier sequence. Every iterate is positive, because a positive t_n gives a positive numerator and denominator. Thus no zero denominator is encountered.

Conclusion. The recursion is Newton's method for $x^2 - 2 = 0$, starting at 1.

31.8 Exercise 31.8: Two real roots of a quartic

What we must show. Count the real roots of $p(x) = x^4 + x^3 - 1$ and approximate each to six decimals.

(a) Establish the shape and count crossings.

The derivative is $p'(x) = x^2(4x + 3)$. It is negative for $x < -3/4$ and positive for $x > -3/4$, except for the isolated zero at $x = 0$. Thus p is strictly decreasing on $(-\infty, -3/4]$ and strictly increasing on $[-3/4, \infty)$. For strict increase across zero, apply the Mean Value theorem separately on either side and combine the inequalities. Now $p(-2) = 7 > 0$, $p(-1) = -1 < 0$, $p(0) = -1 < 0$, and $p(1) = 1 > 0$. The Intermediate Value theorem supplies one root in $(-2, -1)$ and one in $(0, 1)$. Strict monotonicity on the two branches rules out any others.

(b) Run the explicit update.

Newton's formula is

$$x_{n+1} = x_n - \frac{x_n^4 + x_n^3 - 1}{4x_n^3 + 3x_n^2}.$$

Two runs, retaining guard digits, give

n	$x_0 = -1.5$	$x_0 = 0.8$
0	−1.500000000000	0.800000000000
1	−1.398148148148	0.819758064516
2	−1.380747193802	0.819173040661
3	−1.380277903561	0.819172513397
4	−1.380277569098	0.819172513396

As a check independent of apparent convergence, direct substitution gives opposite signs at the endpoints of each interval $(-1.380277570, -1.380277568)$ and $(0.819172512, 0.819172515)$. Each interval contains the unique root on its branch and lies wholly within the rounding range for the answer below.

Conclusion. The two roots, rounded to six decimal places, are −1.380278 and 0.819173.

31.9 Exercise 31.9: Solving the cosine fixed-point equation

What we must show. Approximate the solution of $x = \cos x$ to six decimals. Set $f(x) = x - \cos x$. Since $f'(x) = 1 + \sin x > 0$ on $(0, \pi/2)$, the root there is unique. Newton's iteration is

$$x_{n+1} = x_n - \frac{x_n - \cos x_n}{1 + \sin x_n}.$$

Starting at $x_0 = 1$ gives successively

$$0.750363867840, \quad 0.739112890911, \quad 0.739085133385, \quad 0.739085133215.$$

The signs of f at 0.739085132 and 0.739085135 are negative and positive, respectively. Continuity and strict increase bracket the root, so the displayed digits are not justified solely by two similar iterates.

Conclusion. The root is $x = 0.739085$ to six decimal places.

31.10 Exercise 31.10: Solving an exponential equation

What we must show. Approximate the root of $xe^x = 2$ in $(0, 1)$ to six decimals. Use $f(x) = xe^x - 2$, with $f'(x) = (1 + x)e^x > 0$ on $(0, 1)$. Thus there is at most one root there; the opposite endpoint signs give existence by continuity. Newton's update is

$$x_{n+1} = x_n - \frac{x_n e^{x_n} - 2}{(1 + x_n)e^{x_n}}.$$

Starting with $x_0 = 1$ gives

$$0.867879441171, \quad 0.852783373416, \quad 0.852605526369, \quad 0.852605502014.$$

Direct substitution gives $f(0.852605501) < 0 < f(0.852605504)$, so the unique root is in that small interval.

Conclusion. The six-decimal value is 0.852606.

31.11 Exercise 31.11: A convergent Newton sequence must approach a zero

What we must show. Under the stated bounded-derivative hypotheses, prove $f(x) = 0$ when the Newton iterates converge to the interior point x.

Rearrange the recursion before taking limits.

Since $f'(x_{n-1}) \neq 0$, the update is defined and implies

$$f(x_{n-1}) = (x_{n-1} - x_n)f'(x_{n-1}).$$

Choose $B < \infty$ with $|f'(t)| \leq B$ throughout (a, b). Convergence of (x_n) means both x_n and x_{n-1} tend to x, so their difference tends to zero. Consequently

$$|f(x_{n-1})| \leq B\,|x_{n-1} - x_n| \longrightarrow 0.$$

Identify this limit using continuity.

Differentiability at the interior point x implies continuity there, so also $f(x_{n-1}) \to f(x)$. Uniqueness of a real limit gives $f(x) = 0$. We did not assume f' was continuous or bounded away from zero; the boundedness from above was the estimate actually used.

Conclusion. Every convergent sequence of these iterates with an interior limit converges to a root.

31.12 Exercise 31.12: An interior point is no farther than an endpoint

What we must show. If $a < c < b$, prove $|c - x| \leq \max\{|a - x|\,, |b - x|\}$ for every real x.

Case 1: $x \leq c$**.**

Then $c - x \geq 0$ and $b - x > 0$, so $|c - x| = c - x \leq b - x = |b - x|$.

Case 2: $x > c$**.**

Then $x - c > 0$ and $x - a > 0$, so $|c - x| = x - c \leq x - a = |a - x|$. These cases exhaust the real line. In either case the relevant endpoint distance is at most the maximum of the two endpoint distances.

Conclusion. The claimed inequality follows for all $x \in \mathbb{R}$.